

FFVs can not only reduce CO₂ emissions but also India's dependence on fuel and battery cell imports. Only Ola Electric makes battery cells in India and that too for its own needs. EVs of all major brands have imported battery cells. "For energy resilience and clean mobility, we need to look at multiple sources, as electric vehicles have limitations due to dependence on one country (China). Hence, we are actively focusing on FFVs," says Chitale.

The share of electric in two-

run on up to 100% ethanol on June 4. "There are three pillars which if come together can just help FFVs take off in a big way. The car, the fuel distribution network and the price of ethanol with respect to petrol. The first two are being worked upon and the third is possible at least partially immediately," says Rahul Bharti, Senior Executive Officer, Corporate Affairs, Maruti Suzuki.

Tata Motors Passenger Vehicles is targeting the launch of its maiden FFV by early next

ethanol will be a herculean task.

In its recommendations to the government five years ago, SIAM suggested that E100 can work only when it costs 30% less than gasoline. The industry is asking for policy enablers to encourage consumers to adopt these vehicles, says Toyota's Gulati. "From a customer perspective, the technology has to make sense from the point of view of total cost of ownership, just like in the case of battery electric," he says.

Gulati highlights two immediate challenges: higher upfront cost and loss of mileage. "FFVs cost about ₹50,000 more than a conventional ICE car. With the same GST and road tax rates, the price difference is compounded," he says.

The second challenge is operations. "There is a loss of mileage as ethanol has lower energy density. The price of ethanol must be much less than the price of petrol," says Gulati.

Without a meaningful price advantage over petrol, the trade-off in mileage could make consumers hesitant to switch. "It may end up punishing a consumer for buying a cleaner technology than petrol."

There is also the question of infrastructure such as dispensing machines. The draft CAFE III norms have slashed super-credits on FFVs from 1.5 to 1.1. These remains below strong hybrids, which get 1.6, and electric vehicles, which get 3. Super credits are regulatory multipliers that allow automakers to artificially inflate sales volume of clean, low-emission vehicles when calculating their overall fleet-wide emissions.

"The fleet emission norms must recognise the potential and contribution of ethanol as not only 100% replacement for petrol but also

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wheelers has been stagnant around 6-7% over the past two years. "Our ability to ramp up FFVs is far higher than the ability to penetrate with electric mobility. For EVs, we need to solve many other ecosystem challenges, whereas with ethanol, the industry can move rapidly to FFVs. That is green from the cradle to grave and totally *atmanirbhar*," says Chitale.

For carmakers, too, the rollout timeline is months, says Toyota's Gulati. "That's the time one needs for testing and homologation certification," he says. Toyota is currently running a flex fuel pilot with its strong hybrid multi-purpose vehicle, the Innova Hycross.

Maruti Suzuki India Ltd launched the country's first flex-fuel car—a Wagon R—that can

year. "We are very comfortable in terms of technology readiness. We should be ready with at least one product by the end of this year or early next year," says Shailesh Chandra, MD and CEO, Tata Motors Passenger Vehicles.

Renault, too, is ready to drive in FFVs in the country. "We are fully ready. In Brazil, we have been selling FFVs for over 15 years," says Francisco Hidalgo, vice president of sales and marketing at Renault India.

WILL BUYERS TAKE THE DEAL?

The E20 fuel rollout has already drawn criticism from car owners over lack of an E10 fuel option. Convincing car buyers who are already wary of loss of mileage from